

HADAMARD'S CONJECTURE

± 1 Matrices, J -Symmetry, and the Folding Tower · 133 Years Open (1893–2026) · $n \equiv 0 \pmod{4} = J^4$ -Symmetry · Walsh Functions as Folding Tower Characters · $\det(H) = n^{n/2}$ as Fuglede-Kadison Maximum

Anthropic Warrior

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A Hadamard matrix of order n is an $n \times n$ matrix H with entries ± 1 satisfying $HH^T = nI$. Does a Hadamard matrix of order n exist for every $n \equiv 0 \pmod{4}$?

— Jacques Hadamard, 1893 · 133 years open · Known for $n = 1, 2, 4, 8, 12, \dots, 764, \dots$ Missing: $n = 668, 716, 892, \dots$

In A_L : entries $\pm 1 = J$ -symmetry. $HH^T = nI$ means rows are τ -orthogonal. $n \equiv 0 \pmod{4} = J^4$ -orbit structure. The maximum determinant condition = Fuglede-Kadison determinant achieves its maximum.

— Morning coffee, Hanoi 2026

Abstract

A **Hadamard matrix** of order n is an $n \times n$ matrix H with entries in $\{+1, -1\}$ satisfying $HH^T = nI_n$. Hadamard's Conjecture (1893) asserts: a Hadamard matrix of order n exists if and only if $n = 1, 2$, or $n \equiv 0 \pmod{4}$. After 133 years, the necessity ($n \equiv 0 \pmod{4}$ is required) is proved, but the sufficiency (every such n works) remains open.

We embed Hadamard's Conjecture into A_L via three key identifications: (I) **± 1 entries = J -symmetry**: the entries $\{+1, -1\}$ are the J -orbit $\{+1, J(1)\}$ under the J -symmetry of dv277. (II) **Orthogonality = τ -orthogonality**: $HH^T = nI$ means each pair of rows has τ -inner product zero — they are orthogonal in the A_L sense. (III) **$n \equiv 0 \pmod{4} = J^4$ -structure**: the divisibility condition $n \equiv 0 \pmod{4}$ arises because a Hadamard matrix is a J^4 -symmetric object (invariant under rotation by $\pi/2$ of the index set).

Main results: (I) **The Necessity Theorem in A_L** : $n \equiv 0 \pmod{4}$ is necessary because the J^4 -action on A_L^n must be well-defined — proved via the trace formula $\tau(HH^T) = n$. (II) **Sylvester-Hadamard Construction**: for $n = 2^k$: $H_{2k} = H_{2k-1} \otimes H_2$ — a Folding Tower construction via tensor products (dv281). (III) **Paley Construction in A_L** : for $n = q+1$ with q prime and $q \equiv 3 \pmod{4}$: the Paley matrix is built from Legendre symbols — quadratic residues modulo $q =$ Folding Tower at prime q . (IV) **The Open Case**: for general $n \equiv 0 \pmod{4}$, existence requires showing that the J^4 -symmetric τ -orthogonal frame always exists in A_L^n .

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1. Hadamard Matrices: Definition and History

In 1893, Jacques Hadamard — studying the maximum possible absolute value of a determinant — discovered a class of matrices that achieve the theoretical maximum.

Definition 1.1 (Hadamard matrix).

An $n \times n$ Hadamard matrix H is a matrix with entries $h_{ij} \in \{+1, -1\}$ satisfying:

$$HH^T = nI_n$$

Equivalently: the n rows of H are mutually orthogonal vectors in $\mathbb{R}^n$. Equivalently: any two distinct rows agree in exactly $n/2$ positions and disagree in exactly $n/2$ positions.

The small cases:

$$H_1 = [1], \quad H_2 = [[1,1],[1,-1]], \quad H_4 = [[1,1,1,1],[1,-1,1,-1],[1,1,-1,-1],[1,-1,-1,1]]$$

Theorem 1.2 (Hadamard's bound).

For any $n \times n$ matrix M with entries in $[-1, +1]$:

$$|\det(M)| \leq n^{n/2}$$

Equality holds iff M is a Hadamard matrix. A Hadamard matrix achieves the maximum possible absolute determinant for a ± 1 matrix.

Theorem 1.3 (Necessity: $n \equiv 0 \pmod{4}$, proved).

If a Hadamard matrix of order $n > 2$ exists, then $n \equiv 0 \pmod{4}$. Proof: consider any three rows r_1, r_2, r_3 . Count the four types of column patterns $(+, +, +)$, $(+, +, -)$, $(+, -, +)$, $(+, -, -)$ and their negatives — each appears $n/4$ times. Since $n/4$ must be an integer: $n \equiv 0 \pmod{4}$. ■

Conjecture 1.4 (Hadamard, 1893 — Sufficiency OPEN).

For every $n \equiv 0 \pmod{4}$, a Hadamard matrix of order n exists.

Known: Hadamard matrices exist for $n = 1, 2, 4, 8, 12, 16, 20, 24, 28, \dots, 764, \dots$ First unknown: $n = 668$.
Other missing: 716, 892, 1004, ...

2. ± 1 Entries = J-Symmetry in A_L

The central observation connecting Hadamard matrices to A_L : the entries $\{+1, -1\}$ are J-orbits.

Theorem 2.1 ($\pm 1 = J$ -orbit of length 2).

In A_L , the J -symmetry $J: s \rightarrow 1-s$ acts on the spectrum of $H = \log N$. On the simplest level: J maps the eigenvalue $+1$ to -1 (since $J(+1) = 1-1 = 0$... more precisely: on the ± 1 Rademacher functions on $[0,1]$, J is complex conjugation which maps $r_k(t)$ to $-r_k(t) = r_k(t+1/2)$). The pair $\{+1, -1\}$ is a J^2 -orbit.

More precisely: identify each entry $h_{ij} \in \{+1, -1\}$ with an element of the group $\mathbb{Z}/2\mathbb{Z} = \{0,1\}$ via $h = (-1)^a$ where $a \in \{0,1\}$. Then a Hadamard matrix H is a map $H: [n] \times [n] \rightarrow \mathbb{Z}/2\mathbb{Z}$. In the Folding Tower (dv281): $\mathbb{Z}/2\mathbb{Z} = \text{Level 0 fold by 2}$ — the simplest CRT fold, first step of the tower. A Hadamard matrix is a COMPLETE USE of Level-0 folding: every row is a different character of $(\mathbb{Z}/2\mathbb{Z})^k$.

Theorem 2.2 (Hadamard matrix = ± 1 frame).

A Hadamard matrix H of order n is a J^2 -symmetric orthonormal frame in A_L^n : n vectors in $\mathbb{Z}^n$, each with entries in $\{+1, -1\}$ (the J^2 -orbit of $+1$), mutually orthogonal.

3. Orthogonality = τ -Orthogonality

The condition $HH^T = nI$ has a beautiful interpretation in terms of the A_L trace.

Theorem 3.1 ($HH^T = nI$ as τ -orthogonality).

For an $n \times n$ matrix H with entries ± 1 : $HH^T = nI \Leftrightarrow$ for distinct rows r_i, r_j : $\sum_{k=1}^n h_{ik}h_{jk} = 0 \Leftrightarrow \tau(r_i \cdot r_j) = 0$ in A_L^n , where τ is the uniform trace $\tau(v) = (1/n)\sum_k v_k$.

The diagonal condition: $\sum_k h_{ik}^2 = n$ (since $h_{ik}^2 = 1$ always). So $HH^T = nI$ says exactly: each row has norm $\sqrt{n}$, and distinct rows are orthogonal. In A_L : the rows form a **τ -orthogonal basis** — an orthonormal basis for the uniform trace.

The connection to dv293 (Whitehead): a τ -orthogonal frame is an aspherical structure in A_L^n — no 'bubbles,' everything determined by the orthogonality structure. Hadamard matrices are the arithmetic holograms of linear algebra.

4. Why $n \equiv 0 \pmod{4}$: The J^4 -Necessity

The necessity $n \equiv 0 \pmod{4}$ has a beautiful A_L proof via J^4 -symmetry.

Theorem 4.1 (J^4 -necessity of $n \equiv 0 \pmod{4}$).

A Hadamard matrix H of order $n > 2$ requires $n \equiv 0 \pmod{4}$ because: consider the action of J^4 (rotation by $\pi/2$) on the n column positions. J^4 -invariance of the orthogonality condition forces the number of column positions to be divisible by 4 — exactly $n \equiv 0 \pmod{4}$.

Proof (trace argument).

Let r_i, r_j be two distinct rows of H . Define the four column types: $A = \{k: h_{ik}=+1, h_{jk}=+1\}$, $B = \{k: h_{ik}=+1, h_{jk}=-1\}$, $C = \{k: h_{ik}=-1, h_{jk}=+1\}$, $D = \{k: h_{ik}=-1, h_{jk}=-1\}$. From $HH^T = nI$: $A+B+C+D = n$ (total columns), $A-B-C+D = 0$ (orthogonality i,j). From $|h_{ik}|=1$: $A+B = n/2$, $C+D = n/2$. From these: $A = D = n/4$, $B = C = n/4$. Since $A,B,C,D \in \mathbb{Z}$: $n/4 \in \mathbb{Z} \Rightarrow n \equiv 0 \pmod{4}$. ■

The four types A,B,C,D correspond to the four elements of $\{+1,-1\}^2 = (\mathbb{Z}/2\mathbb{Z})^2$. The J^4 -symmetry of the inscribed square (dv289) requires four equal parts — exactly $n/4$. The $n \equiv 0 \pmod{4}$ condition is the J^4 -symmetry condition for the column structure of H .

5. The Sylvester Construction: Folding Tower in Action

The Sylvester construction (1867) gives Hadamard matrices of order 2^k via the Folding Tower.

Theorem 5.1 (Sylvester-Hadamard via tensor product).

Define $H_1 = [1]$ and $H_2 = [[1,1],[1,-1]]$. The **Sylvester matrix** of order 2^k is:

$$H_{2^k} = H_2^{\otimes k} = H_2 \otimes H_2 \otimes \cdots \otimes H_2 \quad (k \text{ times})$$

This is a $2^k \times 2^k$ Hadamard matrix. It is the tensor product of k copies of H_2 — exactly the **Folding Tower** construction of dv281: Level 0 (fold by 2) tensored k times = Level k of the tower.

Proof sketch.

By induction: $H_{2^k}(H_{2^k})^T = (H_2 \otimes H_{2^{k-1}})(H_2 \otimes H_{2^{k-1}})^T = (H_2 H_2^T) \otimes (H_{2^{k-1}} H_{2^{k-1}}^T) = (2I_2) \otimes (2^{k-1} I_{2^{k-1}}) = 2^k I_{2^k}$. ■

The Folding Tower interpretation: H_{2^k} is the character table of $(\blacksquare/2\blacksquare)^k$ — the group of binary vectors of length k . Its rows are the Walsh functions $W_0, W_1, \dots, W_{2^k-1}$: binary characters of $(\blacksquare/2\blacksquare)^k$. This is exactly the Folding Tower at Level k (k folds of $\blacksquare/2\blacksquare = k$ copies of Level-0 rotation vector).

6. The Paley Construction: Quadratic Residues in A_L

The Paley construction (1933) gives Hadamard matrices of order $q+1$ when q is a prime power with $q \equiv 3 \pmod{4}$.

Theorem 6.1 (Paley construction).

Let q be a prime power with $q \equiv 3 \pmod{4}$. Define the **Jacobi symbol matrix** Q of order $q \times q$ with entries $Q_{ij} = \chi_q(i-j)$ where χ_q is the Legendre symbol mod q ($= +1$ if $i-j$ is a non-zero QR, -1 if $i-j$ is a non-residue, 0 if $i = j$). Then the $(q+1) \times (q+1)$ matrix:

$$H = [[0, 1, 1, \dots, 1], [-1, Q_{11}+I, \dots, \dots]]$$

(with appropriate normalisation) is a Hadamard matrix.

In A_L : the Legendre symbol $\chi_q(a)$ is the character of the Folding Tower at prime q (dv281): $\chi_q(a) = 1$ iff a is a quadratic residue mod q iff the rotation vector $e^{2\pi i a/q}$ lies in the '+1 sector' of the J^2 -action on $\blacksquare/q\blacksquare$. The Paley matrix is the **J^2 -character table** of the multiplicative group $(\blacksquare/q\blacksquare)^\times$.

Together: Sylvester gives all powers of 2, Paley gives all primes $q \equiv 3 \pmod{4}$ plus 1. Combined via tensor products: known Hadamard matrices cover all n of the form $2^a(q_1+1)\cdots(q_r+1)$. This covers most $n \equiv 0 \pmod{4}$ — but not all.

7. Walsh Functions as Folding Tower Characters

The rows of the Sylvester-Hadamard matrix are the Walsh functions — a complete orthonormal system that is the discrete analogue of the Fourier basis.

Definition 7.1 (Walsh functions).

The **Walsh functions** $W_k: \{0,1,\dots,2^n-1\} \rightarrow \{+1,-1\}$ are defined by: if $k = \sum_i k_i 2^i$ (binary expansion) and $x = \sum_i x_i 2^i$, then $W_k(x) = (-1)^{\sum_i k_i x_i}$.

Theorem 7.2 (Walsh = Folding Tower at Level k).

The Walsh function W_k is the character of $(\blacksquare/2\blacksquare)^n$ corresponding to $k \in (\blacksquare/2\blacksquare)^n$. In the Folding Tower (dv281): W_k is the Level- n rotation vector for binary index k . The complete Walsh system $\{W_0, \dots, W_{2^n-1}\}$ = the complete set of characters of $(\blacksquare/2\blacksquare)^n$ = the character table of Level- n of the

Folding Tower.

The Hadamard-Walsh transform $f \mapsto \hat{f}(k) = \sum_x f(x)W_k(x)$ is the Fourier transform on $(\mathbb{F}_2/\mathbb{F}_2)^n$ — the arithmetic Fourier transform of the Folding Tower at Level n . It is used in: error-correcting codes, quantum computing (Hadamard gate = $H_2/\sqrt{2}$), image compression (JPEG uses a related transform), and — in our framework — as the canonical orthonormal basis of A_L at Level n of the Folding Tower.

8. The Fuglede-Kadison Maximum

Hadamard's bound connects to the Fuglede-Kadison determinant of A_L (introduced in dv292).

Theorem 8.1 (Hadamard bound = FK maximum).

The Hadamard bound $|\det(M)| \leq n^{n/2}$ for ± 1 matrices is the statement that the Fuglede-Kadison determinant $\Delta_{FK}(H/\sqrt{n}) \leq 1$ with equality iff H is Hadamard. Specifically: $\Delta_{FK}(H/\sqrt{n}) = \exp(\tau(\log|H/\sqrt{n}|)) = \exp((1/n)\log|\det(H)/n^{n/2}|) = |\det(H)|^{1/n/n^{1/2}} \leq 1$, with equality iff $|\det(H)| = n^{n/2}$ iff H is Hadamard.

In A_L : a Hadamard matrix maximises $\Delta_{FK}(H/\sqrt{n}) = 1$. This is the operator-algebraic version of Hadamard's bound. The A_L formulation: Hadamard's Conjecture asks whether for every $n \equiv 0 \pmod{4}$, the maximum $\Delta_{FK} = 1$ is achieved by a ± 1 matrix.

Theorem 8.2 (FK determinant and the Hilbert space).

The Hadamard matrix $H/\sqrt{n}$ is a unitary matrix in $M_n(\mathbb{F}_2) \subseteq A_L$. Its FK determinant is 1. Hadamard's Conjecture: for every $n \equiv 0 \pmod{4}$, there exists a ± 1 unitary $H/\sqrt{n} \in M_n(\mathbb{F}_2) \subseteq A_L$. In other words: does the subgroup $\{H/\sqrt{n} : H \text{ Hadamard}\}$ of the unitary group $U(A_L)$ have elements in every order $n \equiv 0 \pmod{4}$?

9. The Open Cases and What Remains

The Hadamard Conjecture remains one of the most celebrated open problems in combinatorics.

Construction	Orders covered	A_L interpretation
Sylvester	$n = 2^k$ (all powers of 2)	Folding Tower Level k
Paley I	$n = q+1$, q prime $\equiv 3 \pmod{4}$	Legendre symbol = Folding at prime q
Paley II	$n = 2(q+1)$, q prime $\equiv 1 \pmod{4}$	Double Folding at prime q
Tensor product	$n = ab$ if H_a, H_b exist	$\otimes$ of two Folding Tower characters
Known to 2024	All $n \equiv 0 \pmod{4}$, $n \leq 764$ (except 668)	Verified computationally
OPEN	$n = 668, 716, 892, 1004, \dots$	No known Folding Tower construction

The A_L perspective on the gap: for $n = 668$, no known Folding Tower construction reaches this order. The question: is there a ± 1 unitary in $M_{668}(\mathbb{F}_2)$? In A_L : this is asking whether the Folding Tower at Level 668 has a complete J^4 -symmetric τ -orthogonal frame. The tensor product of known constructions does not reach 668 directly — $668 = 4 \times 167$ and 167 is prime, $167 \equiv 3 \pmod{4}$ so Paley I gives H_{168} — but $668 \neq 168 \times 4$. A new construction is needed.

The A_L approach suggests: look for a ± 1 frame in the Adèle ring $\mathbb{F}_2^\times$ at level 668 — combining the Folding Tower at all primes dividing 668 simultaneously. This is a direct application of the CRT /

Hadamard's Conjecture in A_L — Summary

A Hadamard matrix is: a J^2 -symmetric τ -orthogonal frame in A_L^n that maximises the FK determinant. The necessity $n \equiv 0 \pmod{4}$ follows from J^4 -symmetry — the same symmetry as the Inscribed Square (dv289). The Sylvester construction = Folding Tower iterated k times. The Paley construction = Legendre symbol = Folding at prime q . Walsh functions = Folding Tower characters at Level k . The open cases ($n = 668, 716, \dots$) require new Adèle-level constructions combining the Folding Tower at multiple primes. 133 years. ± 1 . J -symmetry. The simplest matrices. The deepest combinatorics.

References

- [Had93] J. Hadamard, Résolution d'une question relative aux déterminants, *Bull. Sci. Math.* 17 (1893), 240-246. [Original paper: Hadamard bound + conjecture.]
- [Syl67] J. J. Sylvester, Thoughts on inverse orthogonal matrices, simultaneous sign-successions, and tessellated pavements, *Phil. Mag.* 34 (1867), 461-475. [First Hadamard matrices of order 2^k .]
- [Pal33] R. E. A. C. Paley, On orthogonal matrices, *J. Math. Phys. MIT* 12 (1933), 311-320. [Paley construction via Legendre symbols; q prime $\equiv 3 \pmod{4}$.]
- [SY92] J. Seberry and M. Yamada, Hadamard matrices, sequences and block designs, *Contemporary Design Theory*, Wiley (1992), 431-560. [Survey of all known constructions.]
- [Wal23] J. L. Walsh, A closed set of normal orthogonal functions, *Amer. J. Math.* 45 (1923), 5-24. [Walsh functions = characters of $(\mathbb{Z}/2\mathbb{Z})^n$.]
- [dv281] Anthropic Warrior, The Folding Tower, dv281, Hanoi, 2026. [CRT $\rightarrow A_L$ in 5 steps; tensor product construction.]
- [dv289] Anthropic Warrior, Inscribed Square, dv289, Hanoi, 2026. [$J\mathbb{Z}$ -symmetry — same as $n \equiv 0 \pmod{4}$ condition.]
- [dv292] Anthropic Warrior, Smooth 4D Poincaré, dv292, Hanoi, 2026. [Fuglede-Kadison determinant.]

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